Halves and Fourths — Are the Parts Really Equal? Answer Key

Kindergarten–Grade 1

Quick Answers

1. —	3. 2	5. 5
2. —	4. 2, 2, 6, 6	6. —

Curriculum

Level: Kindergarten–Grade 1
1.G – *problems 1, 2, 3, 4, 5, 6*
Difficulty 1–5 of 5 across 6 tagged checks.

Common wrong answers

3. *If they got 4: moved the whole difference across, which only swaps which part is bigger*
5. *If they got 10: folded once and shared between two parts instead of four*

1. Warm up. Rosa's rectangle is cut into Part A and Part B. Are the two parts the same size?

Step 1: name what makes a half — two parts are halves only when they hold the same number of little squares, so the job is to count, not to look.
Step 2: count Part A. It is two columns of three squares: 3 and 3 more makes 6 squares.
Step 3: count Part B the same way: 6 squares.
Step 4: the two counts match, so the correct sign between them is "equals": $\boxed{6 = 6}$
Both parts are halves. *Grading note:* accept a drawn = sign, the word "same", or "equal".

2. Kim says Part A (3 squares) is one half of a rectangle whose other part, Part B, has 9 squares. Write the true comparison.

Step 1: find Kim's mistake — Kim looked at the *number of parts* (two parts) instead of the *size* of the parts. Cutting a shape into two pieces does not make halves unless the pieces match.
Step 2: count each part: Part A holds 3 squares, Part B holds 9 squares.
Step 3: compare the counts. 3 is a smaller count than 9, so Part A is the smaller piece: $\boxed{3 < 9}$
Step 4: say why — because 3 is not the same as 9, Part A is not one half. It is much smaller than half.

3. Ben has parts of 4 and 8 squares and wants to move squares until the parts are halves. He says to move 4. How many squares should really move?

Step 1: find the fair share first. Ben's whole rectangle holds 4 and 8 squares together, which is 12 squares, and two equal halves must hold the same amount, so each half needs 6 squares.
Step 2: Part A already holds 4 squares, so it needs $6-4$, which is 2 more squares.
Step 3: check Ben's move. If 4 squares moved across, Part A would have 8 and Part B would have 4 — the parts would just trade places, and they still would not match. Ben moved the whole *difference* instead of half of it.
Step 4: move this many squares:

$$\boxed{2}$$

Check: $4+2=6$ and $8-2=6$ ✓

4. Kai's four pieces cover 2, 6, 6, and 2 squares. Write the four counts in order, smallest first.

Step 1: read the four counts off the picture: Piece A = 2, Piece B = 6, Piece C = 6, Piece D = 2.
Step 2: put the smallest first. The two small pieces each hold 2 squares and the two big pieces each hold 6 squares, so the order is:

$$\boxed{2, 2, 6, 6}$$

Step 3: read what the order shows — the counts are not all the same, so Kai's pieces are *not* fourths. Four pieces alone is not enough; all four must match.

5. Ivy's rectangle has 20 squares. She folded it once, counted one of the two parts, and wrote 10 squares for a fourth. How many squares should each fourth really have?

Step 1: find Ivy's mistake — folding one time makes *two* parts, not four. The number she counted, 10, is one **half** of the rectangle.
Step 2: fourths means sharing the whole rectangle equally among 4 parts, so share the 20 squares into 4 equal groups.
Step 3: 20 shared into 4 equal groups gives:

$$\boxed{5}$$

Step 4: check by putting the fourths back together: $5+5+5+5=20$ ✓ (and folding a second time turns each half of 10 into two fourths of 5).

6. Challenge (open response — review by hand). Draw one straight line that cuts a rectangle of twelve squares into halves, and explain how the squares show the parts are halves.

Model answer. One straight cut down the middle of the long side makes two parts of six squares each. A cut across the middle of the short side also works, and so does a cut along a diagonal of the whole rectangle.
What a full answer says: "I counted six squares on this side and six squares on that side. The counts are the same, so the two parts are halves."
Look for: the line makes exactly two parts; the two parts hold the same number of squares; and the explanation talks about the parts being the **same size**, not just about there being two parts. A child who says "it is halves because there are two pieces" has the misconception this sheet targets — send them back to count.